inspect_model.py

`scripts/inspect_model.py · 114 lines`

What it does

First stage. Loads the configured model, lists every sparse MoE layer and writes a machine-readable architecture report. A probe forward pass verifies that each layer's output equals the sum of gate times expert output, i.e. where and how often the gate is applied.

Run

```
uv run python scripts/inspect_model.py [--profile ...]
```

Pipeline order: inspect_model -> build_c4_support_eval -> collect_stats -> score_experts -> replace_experts -> evaluate (before_training) -> train_biases -> evaluate (after_training) -> export_model

Reads

- the configured model (`HF_MODEL_ID` / `MODEL_PATH`)

Writes

- `<run_dir>/architecture_report.json`

Notes

- Exits with an error if the gate check fails or if float16 produces inf/NaN.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`main()`	Script entry point: runs this stage end to end.

Connections

Uses: `moe_pruning.data.calibration`, `moe_pruning.model.architecture`, `moe_pruning.model.instrumentation`, `moe_pruning.model.loader`, `moe_pruning.utils.cli`, `moe_pruning.utils.paths`, `moe_pruning.utils.reproducibility`